

What is CPM?

The critical path method (CPM) is a step-by-step technique for scheduling project activities.

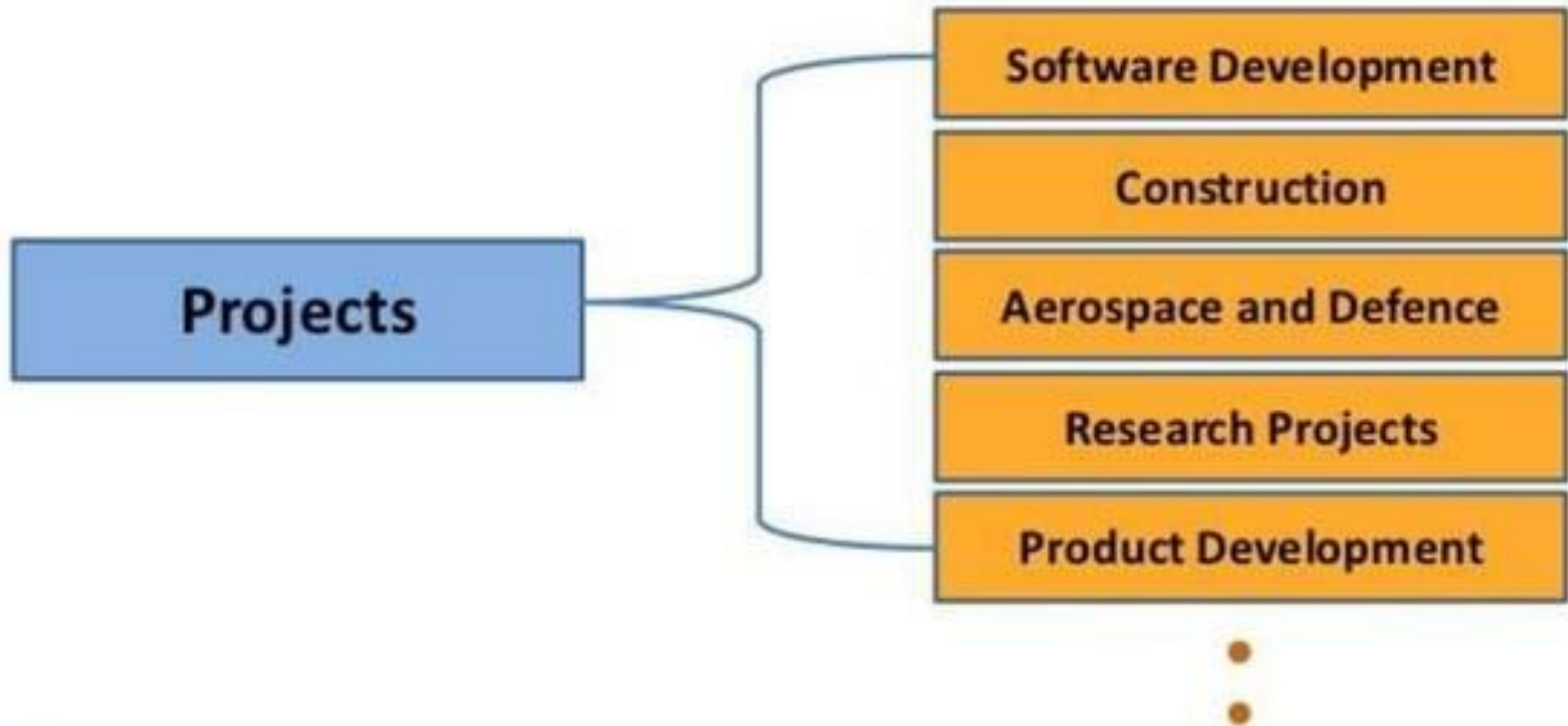


Critical Path Analysis

- **Critical Path Analysis (CPA)**, is sometimes called Network Analysis
- It is a tool used to plan activities so that a job can be completed in the shortest time
- It breaks a job down into a number of tasks, and looks at the dependency of them
 - For example, list the activities that must be completed in order to make a cup of coffee
- It is used commonly in manufacturing and construction



Where is it used ?



NASA used the critical path method in 1969 to help determine an efficient schedule for the tasks that led to **the first moon landing**.

Example : Redesign of a Product

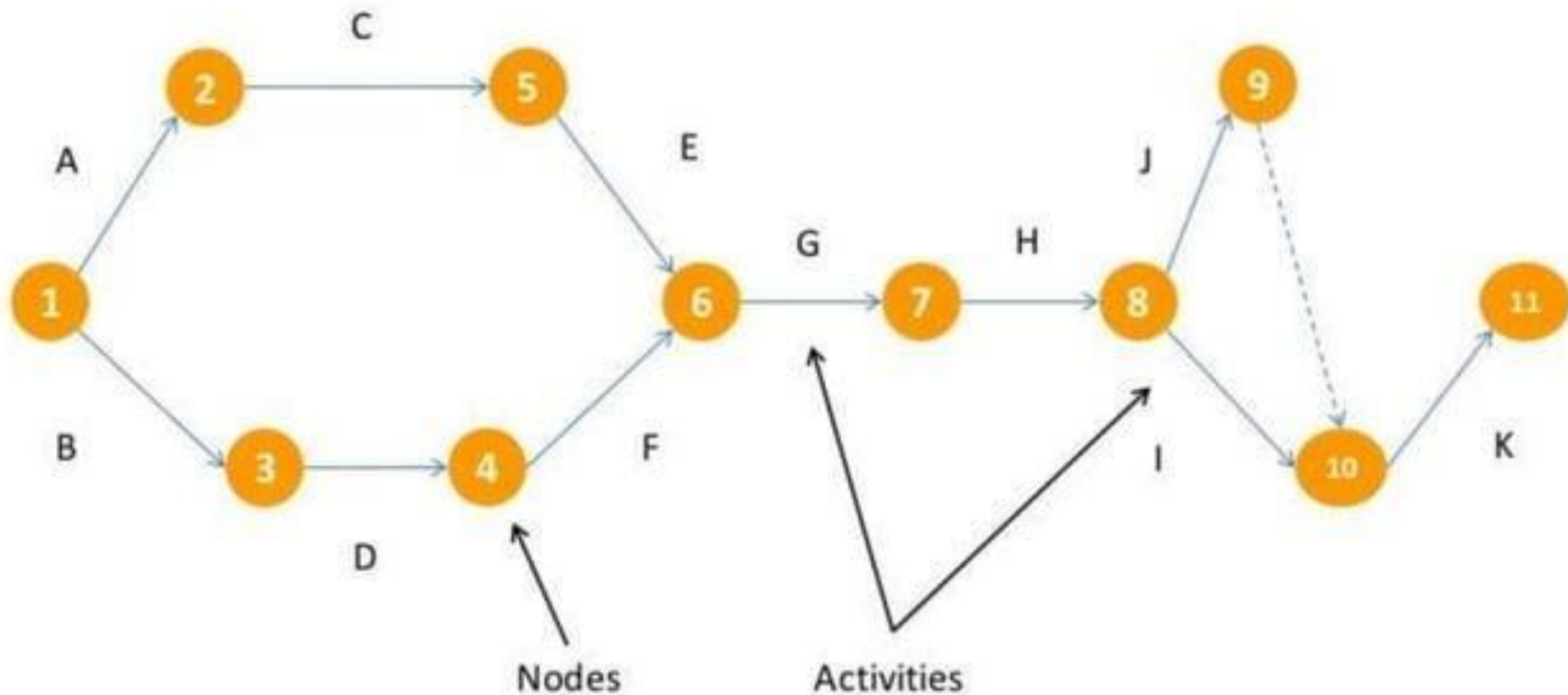
Activity	Description	Predecessor	Duration (weeks)
A	Redesign Product	-	6
B	Redesign Packaging	-	2
C	Order / Receive components for redesigned product	A	3
D	Order / Receive components for redesigned packaging	B	2
E	Assemble products	C	4
F	Make up packaging	D	1
G	Package the product	E,F	1
H	Test market the packaged product	G	6
I	Revise redesigned product	H	3
J	Revise redesigned packaging	H	1
K	Prepare report	I,J	1

Step 1 – Identify Paths

Activity	Predecessor	Duration (weeks)
A	-	6
B	-	2
C	A	3
D	B	2
E	C	4
F	D	1
G	E,F	1
H	G	6
I	H	3
J	H	1
K	I,J	1

A-C-E-G-H-I-K
A-C-E-G-H-J-K
B-D-F-G-H-I-K
B-D-F-G-H-J-K

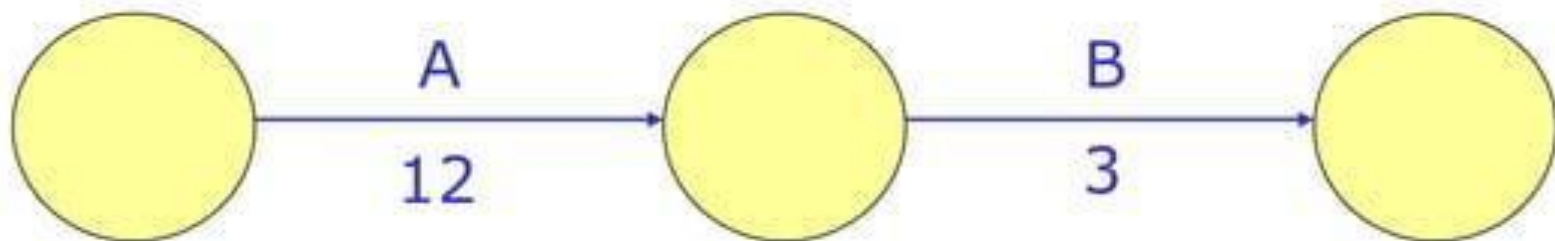
Step 2 – Network Diagram





A Simple Network Diagram

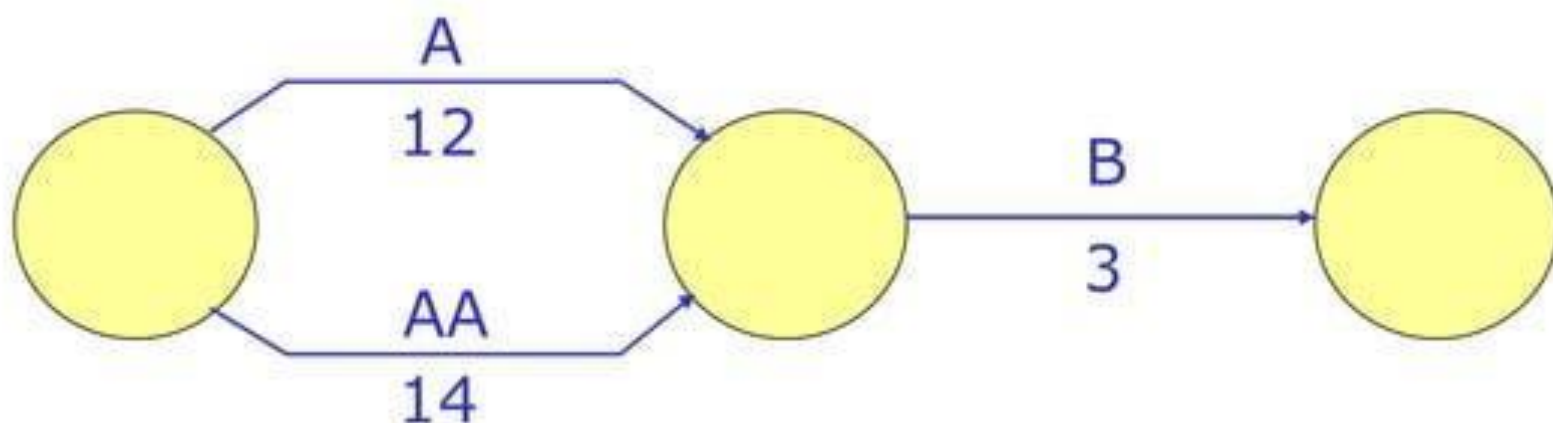
- A business wishes to build a new factory
- Before it can do so it needs to:
 - Buy the land (Activity A – will take 12 weeks)
 - Draw up Plans (Activity B – will take 3 weeks)
- A simple network may be drawn to illustrate this scenario:





Illustrating Simultaneous Activities

- Of course in reality some activities can be carried out simultaneously
- Using the previous example:
 - Assume that whilst in the process of buying the land the firm wants to apply for planning permission (Activity AA – will take 14 weeks)



- Try drawing a network using your instructions for making a cup of coffee



Calculating The Earliest Start Time

- The main reason for drawing a network is to identify the **CRITICAL** activities
- To do this we must calculate the earliest time at which any given activity can start
 - This is called the **Earliest Start Time (EST)** of the activity
- It is calculated using the following formula:

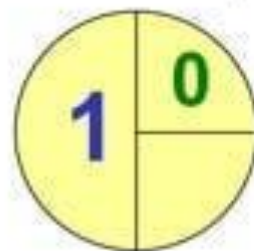
$$\text{EST} = \text{EST of Previous activity} + \text{Duration of previous activity}$$



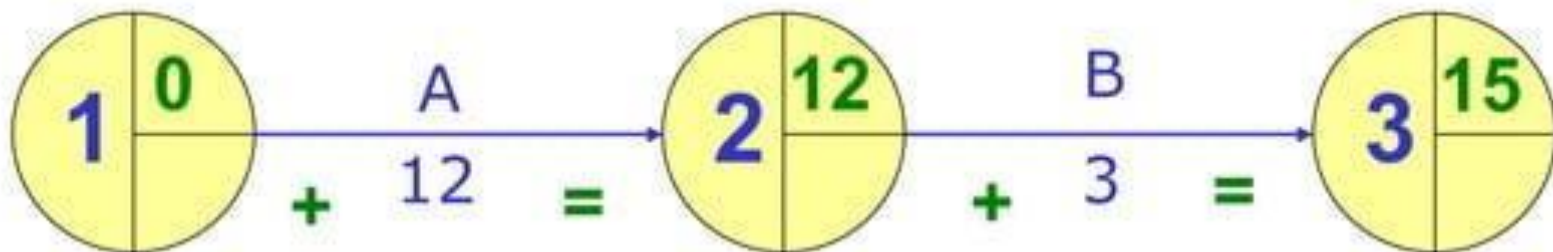


Calculating The Earliest Start Time

- This information is then placed in the top right-hand quarter of the node
- The first node will **ALWAYS** have an EST of zero
- Using our first simple example, the EST for each activity would be calculated as follows:



EST goes in
top right-
hand
quarter of
the node

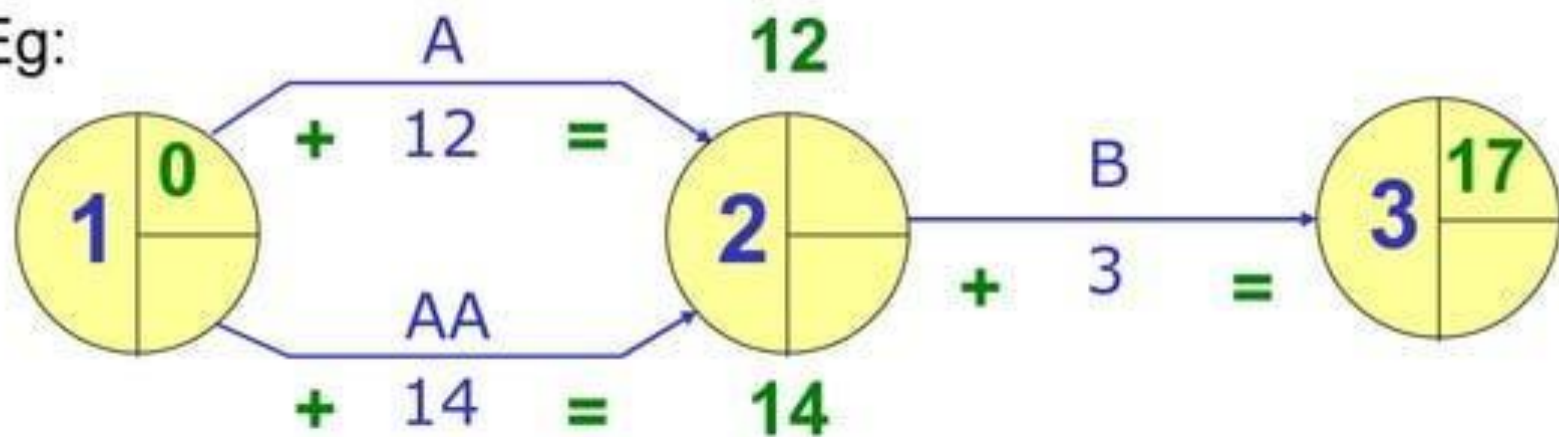




The EST and Simultaneous Activities

- When there are simultaneous activities there may be more than one value for the EST

Eg:



- Since Activity B is **DEPENDENT** on both Activities A and AA, it cannot start until both are complete
- So we must take the **HIGHEST** figure
- This means that the EST is 14 weeks



Recap of The Earliest Start Time

- The EST of the first activity is always zero
- Calculate the EST by working left to right across a network
- It is calculated using the following formula:

$$\text{EST} = \text{EST of Previous activity} + \text{Duration of previous activity}$$

- When there are 2 simultaneous activities the **HIGHEST** figure is used as the EST





Calculating The Latest Finishing Time

- There is one final piece of information needed to complete our network diagram
- To identify the **CRITICAL** activities we must also know the latest time at which any given activity must end
 - This is called the **Latest Finishing Time (LFT)** of the activity
- It is calculated by working **BACKWARDS** across the network using the following formula:

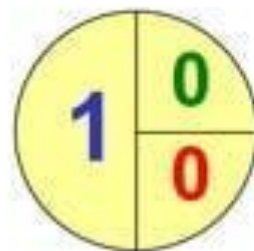
$LFT = LFT \text{ at end of following activity} - \text{Duration of following activity}$



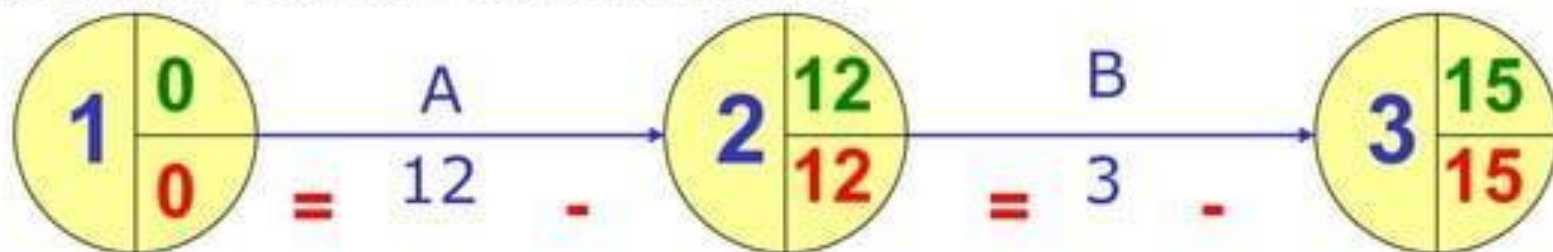


Calculating The Latest Finishing Time

- This information is then placed in the bottom right-hand quarter of the node
- The first node will **ALWAYS** have an LFT of zero
- Using our first simple example, the LFT for each activity would be calculated as follows:



LFT goes in
bottom
right-hand
quarter of
node



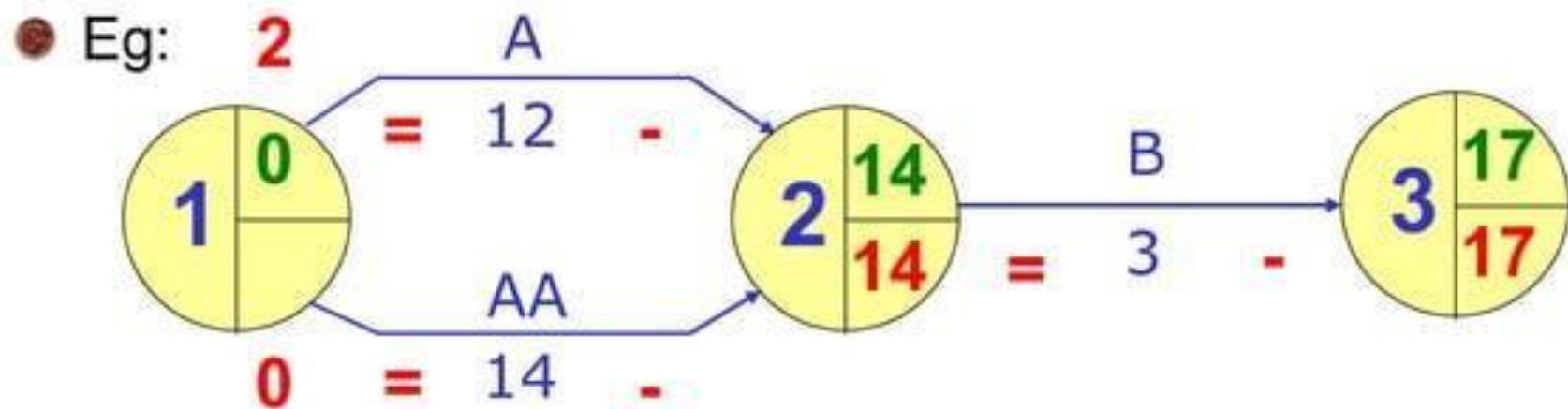
Since the earliest time this project can be finished is 15 weeks then this is also the latest we would like to finish the project. As such:

The EST and LFT of the last node are ALWAYS the same



The LFT and Simultaneous Activities

- When there are simultaneous activities there may be more than one value for the LFT



- If Activity AA starts on week 2 it cannot be completed by week 14
- So we must take the **LOWEST** figure
- This means that the LFT is week 0



Recap of The Latest Finishing Time

- The LFT of the last activity is always equal to its EST
- The LFT of the first activity is always zero
- Calculate the LFT by working right to left across a network
- It is calculated using the following formula:

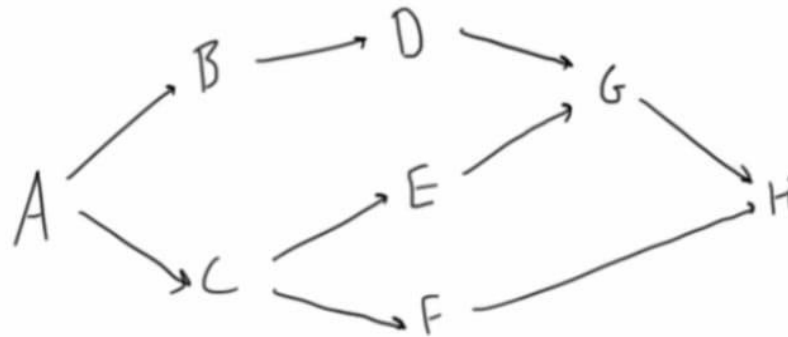
$$\text{LFT} = \text{LFT at end of following activity} - \text{Duration of following activity}$$

- When there are 2 simultaneous activities the **LOWEST** figure is used as the LFT

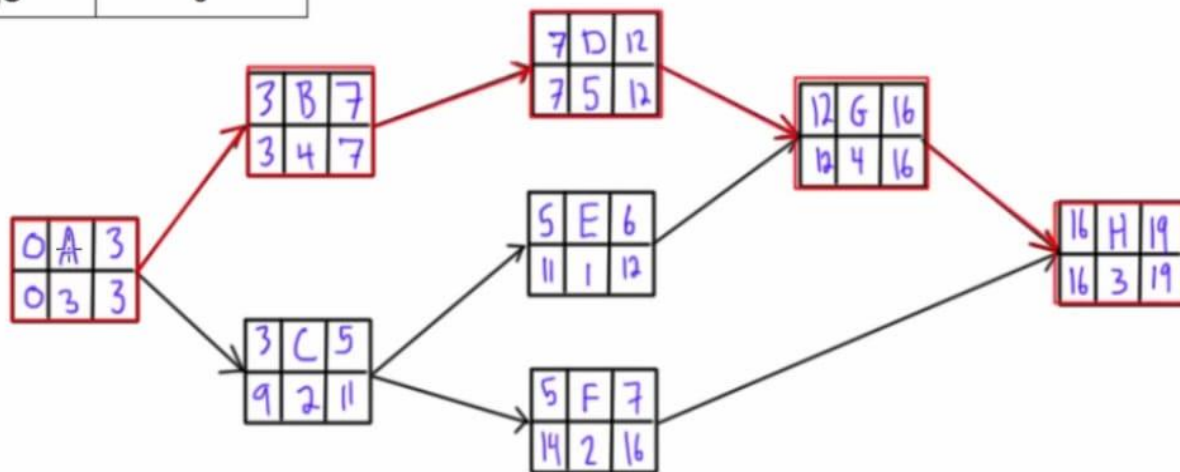


PROBLEM SOLVING ON CPA

Activity	Predecessor	Duration (days)
A	-	3
B	A	4
C	A	2
D	B	5
E	C	1
F	C	2
G	D,E	4
H	F,G	3

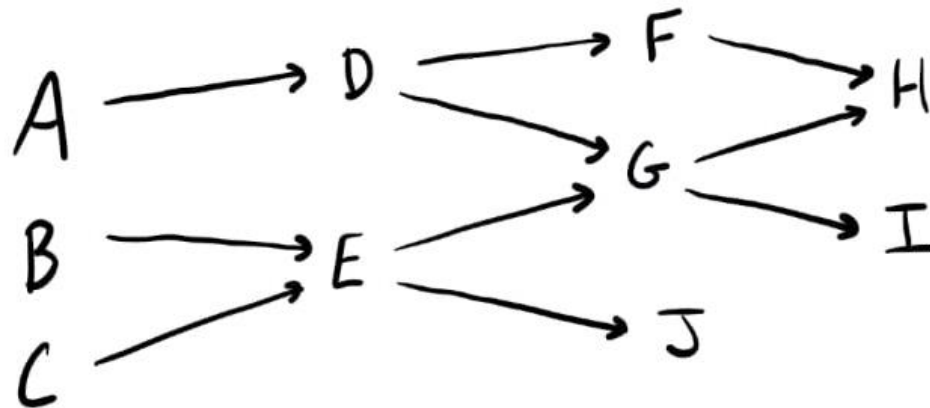


ES	Act	EF
LS	dur	LF

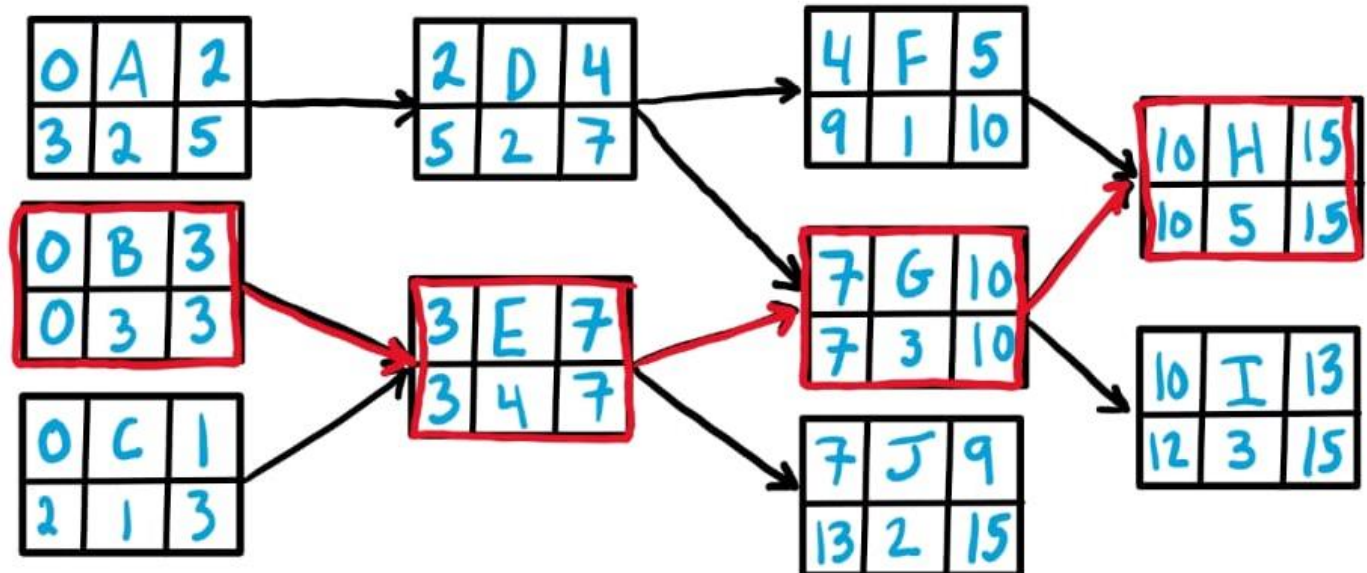


PROBLEM SOLVING ON CPA

Activity	Predecessor	Duration (days)
A	-	2
B	-	3
C	-	1
D	A	2
E	B,C	4
F	D	1
G	D,E	3
H	F,G	5
I	G	3
J	E	2



ES	Act	EF
LS	dur	LF



Exercises on CPA

Considering the project information presented in the table below of Critical Path Analysis

Activity	Predecessores	Duration (Days)
A	—	10
B	A	8
C	A	25
D	B	10
E	C	3
F	C	4
G	D, E	15
H	F, G	4

Develop the network diagram from the above table and **discover** the total duration of the project. Also, show all the possible paths and their durations, and **identify** the critical path of the project.

Exercises on CPA

Considering the project information presented in the table below of Critical Path Analysis

Activity	Immediate Predecessor	Time (days)
A	-	2
B	-	5
C	-	1
D	B	10
E	A, D	3
F	C	6
G	E, F	8

Visualize the network diagram from the above table. Also, **estimate** all the possible path and their duration and **compute** the critical path of the project.

Exercises on CPA

Consider the following project schedule with activity labels, durations (in weeks), and their respective precedents:

Activity	Duration (weeks)	Precedents
A	7	-
B	5	-
C	3	A
D	6	B
E	4	C, D
F	9	E
G	4	E
H	3	F, G
I	1	H

Convert the given table into a Network Diagram and find the critical path showing the earliest start time (ES), earliest finish time (EF), latest start time (LS), latest finish time (LF).

Exercises on CPA

Consider the following project schedule with activity labels, durations (in weeks), and their respective precedents:

Activity	Duration (weeks)	Precedents
A	7	-
B	5	-
C	3	A
D	6	A, B
E	4	B, C
F	9	A
G	4	E, F
H	3	D, G

Apply the critical path method to find the critical path and the activities involved. **Estimate** the earliest start time (ES), earliest finish time (EF), latest start time (LS), latest finish time (LF), and also identify the critical path.